DEEPCORE LABS · OPENEGIZ DIGITAL MINE 2026

MineGuard X

Physics-informed underground mine digital twin

Connected methane, ventilation, uncertainty, simulated safety, and energy decisions across six fictional zones.

SIMULATION ONLY

OPERATIONAL QUESTION

One system, interacting constraints

Methane transport

A reading depends on sources, airflow, volume, and upstream concentration.

Telemetry uncertainty

Stale or invalid sensors become UNKNOWN_DATA, never a safe value.

Ventilation reliability

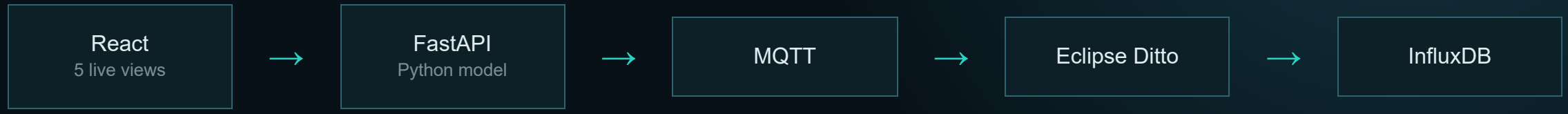
Fan failure changes airflow and takes precedence in the risk model.

Energy

Optimization is allowed only after configured safety constraints are met.

ARCHITECTURE

Official OpenEgiz path, preserved



Two nonroot Compose extension services reuse official networking, credentials, initialization, and telemetry infrastructure. Platform I/O cannot freeze the deterministic physics clock.

PHYSICS MODEL

Conservative, deterministic, explicit units

```
Z01 ? Z02 ? [Z03 ? Z04] ? Z05 ? Z06
```

Z02 splits into two parallel working branches; both join at Z05. CO uses a separate conservative tracer.

$$V\,dc/dt = \Sigma(q_{in}\,c_{in}) - \Sigma(q_{out}\,c) + source$$

$$\Delta p = \Delta p_{shutoff}\,s^2 - Kq^2 \quad \cdot \quad P = P_{rated}\,s^3$$

Methane is an internal volume fraction and displayed as % volume. All geometry and parameters are declared fictional, uncalibrated assumptions.

RISK AND SIMULATED SAFETY

Uncertainty and safety precede optimization

NORMAL → ADVISORY → WARNING → EMERGENCY

Configured demonstration thresholds use downgrade hysteresis and reason codes. They are not statutory mine limits.

UNKNOWN_DATA

Stale/invalid telemetry disables optimization and triggers conservative simulated guidance.

Fan failure precedence

A failed main fan creates EMERGENCY regardless of the current methane reading.

Simulated actions

Production stop and evacuation stay simulated. Healthy fan models achieve emergency requests; failed fans stay excluded.

VENTILATION ON DEMAND

Feasibility first, energy second

1. Validate risk and data quality
2. Enumerate ramp-bounded fan settings
3. Forecast coupled airflow and methane
4. Reject every constraint violation
5. Select minimum energy + switching penalty

Configured initial-state model estimate

1.171 kWh baseline

0.599 kWh selected

48.8% calculated difference

Uncalibrated synthetic 60-second model estimate. Not measured industrial savings.

SCENARIOS AND EVIDENCE

Reproducible behavior, visible causes

S1 · Methane release

Declared Z04 source increase propagates through gas transport, risk, safety, and Ditto state.

S2 · Main fan failure

Failure at T+10 s produces zero network flow and emergency precedence.

Pause/resume preserves simulation time. Reset restores the exact versioned baseline. Command Center shows requested versus actual fan feedback. Energy shows feasible/rejected forecasts. S1 reached emergency at T+42 s and 1.5072% methane; Ditto interlock and reset were verified.

LIVE DITTO READBACK VERIFIED

REPRODUCE

One startup command

```
bash mineguard/scripts/bootstrap.sh
```

Open `http://127.0.0.1:8091`

```
bash mineguard/scripts/full-smoke.sh
```

Local gate: 194 backend tests, 32 frontend tests, official smoke and Ditto readback. Desktop/mobile browser routes passed. Clean-VM rehearsal and final submission-branch verification remain pending.

FICTIONAL MODEL · SIMULATED ACTIONS · OPENEGIZ INTEGRATED